Benefits of Conservation Easements to Biodiversity: Evidence from Wetland Birds

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Background - Wetland at Risk

- 2019: wetlands (95% freshwater) cover under 6% of the lower 48 states — half their historical extent
  - Loss risen $\sim$ 50% since 2009, threatening water quality, flood protection, and wildlife habitat (USFWS, 2024).
- Wetlands: provide habitat provision for many different bird species and deliver essential ecosystem services (Bolduc & Afton, 2008; McKinney et al., 2011).
- **"Waters of the United States (WOTUS)" protections: many wetlands outside federal jurisdiction** (Greenhill et al., 2024)
  - Reinforcing the role of conservation efforts (**easements**).
  - Historical policy regimes suggest plausible exogenous variation for instrument.

Background - Conservation Easement

- Conservation easement: voluntary, legally binding agreement limiting land uses to protect conservation values - land in private ownership
  - landowners give up development rights for compensation or tax benefits
  - USDA's NRCS has obligated over $1.8 billion to preserve nearly 800,000 acres through agricultural and wetland easements + 2018 Farm Bill allocates approximately $450 million annually to ACEP
- Easements are perpetual property interests that run with the land rather than expiring with a contract.

Motivation

- Literature highlights the role of conservation practices - wetland protection - in improving natural resources (Prokopy et al., 2017; Liu et al., 2023; Skidmore et al., 2023; Zhang et al., 2025).
- Birds offer a direct, responsive indicator of conservation impact (easily observable and respond more timely to habitat changes than vegetation or nutrient conditions) (Haslem et al., 2023).

→ Causal effects of conservation easements on biodiversity are relatively underexplored - crucial for linking long-term conservation commitments to measurable benefits to ecosystem services.

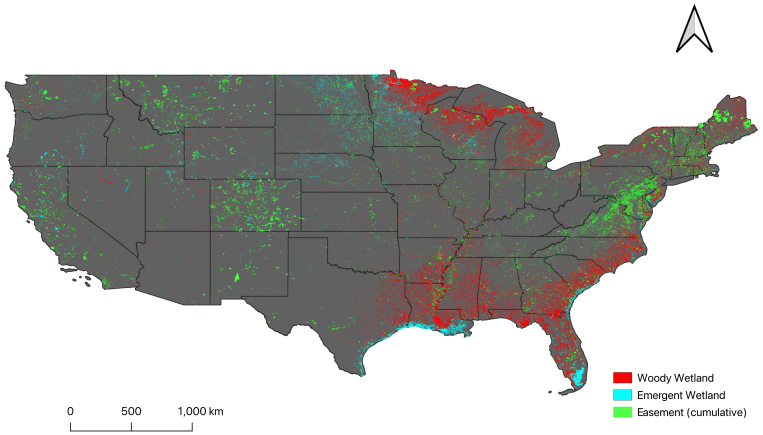
Research Question

- **What are the ecological benefits of conservation easements established around wetlands for bird populations**?
  - Estimate the causal effects of easements on bird species relative abundance and richness indices
  - What is the economic value of biodiversity gains generated by conservation easements on wetlands?

Data

1. Conservation Easement
   - Easement-level acreage with unique identifier
   - Date of establishment, owner types and conservation purpose
2. eBird
   - Checklist-level observational data
   - Hour-day-year timestamp, lat.-long., unique ID
3. Land Cover
   - 30-meter spatial resolution
   - Wetland class: Woody Wetlands + Emergent Herbaceous Wetlands
4. Ancillary Data
   - Watershed boundary (HUC)
   - Controls: Weather, ag land values

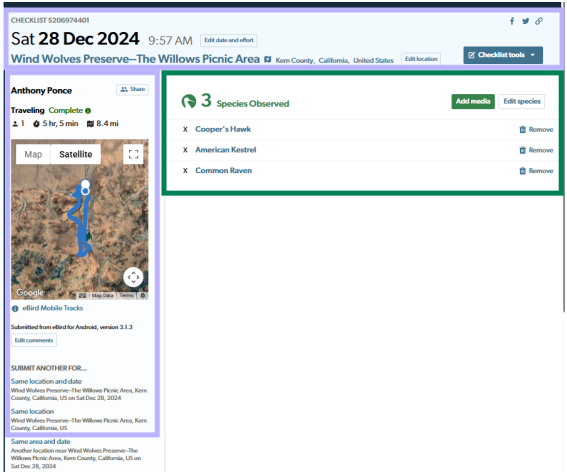
Conservation Easements and Wetlands

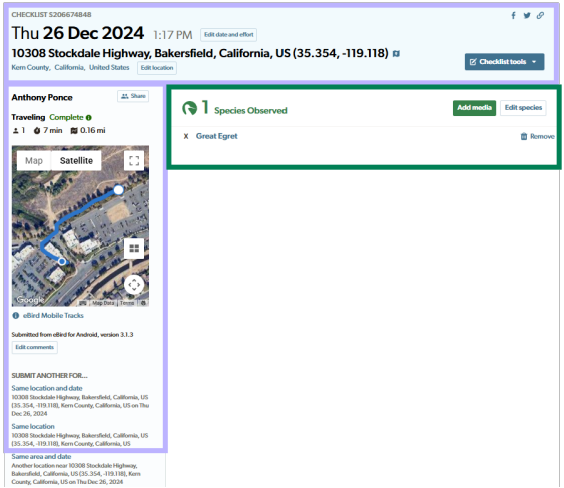


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eBird Data

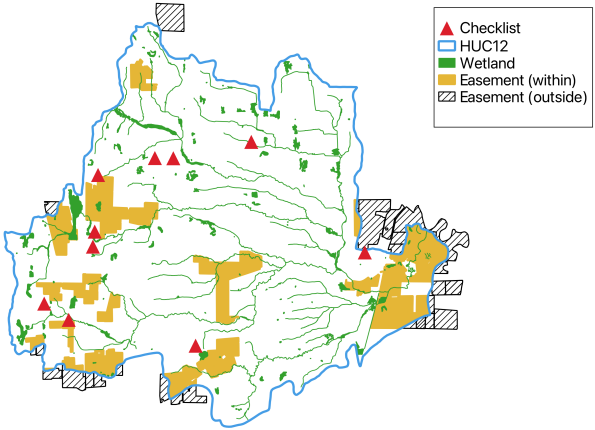

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Kern County, California, United States
Edit location
Checklist tools
Anthony Ponce
Share
Traveling Complete
1 5 hr, 5 min 8.4 mi
Map Satellite
eBird Mobile Tracks
Submitted from eBird for Android, version 3.1.3
Edit comments
SUBMIT ANOTHER FOR...
Same location and date
Wind Wolves Preserve--The Willows Picnic Area, Kern County, California, US on Sat Dec 28, 2024
Same location
Wind Wolves Preserve--The Willows Picnic Area, Kern County, California, US
Same area and date
Another location near Wind Wolves Preserve--The Willows Picnic Area, Kern County, California, US on Sat Dec 28, 2024
3 Species Observed
Add media
Edit species
x Cooper's Hawk
Remove
x American Kestrel
Remove
x Common Raven
Remove


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Kern County, California, United States
Edit location
Checklist tools
Anthony Ponce
Share
Traveling Complete
1 7 min 0.16 mi
Map Satellite
eBird Mobile Tracks
Submitted from eBird for Android, version 3.1.3
Edit comments
SUBMIT ANOTHER FOR...
Same location and date
10308 Stockdale Highway, Bakersfield, California, US (35.354, -119.118), Kern County, California, US on Thu Dec 26, 2024
Same location
10308 Stockdale Highway, Bakersfield, California, US (35.354, -119.118), Kern County, California, US
Same area and date
Another location near 10308 Stockdale Highway, Bakersfield, California, US (35.354, -119.118), Kern County, California, US on Thu Dec 26, 2024
1 Species Observed
Add media
Edit species
x Great Egret
Remove

Data

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Matching using National Land Cover Database + NHDPlus



Identification

Estimating Relative Bird Abundance and Richness

Poisson process (Log-Linear) to estimate bird abundance conditional on checklist characteristics (Liang et al.,2020)

$$\mathrm{B}_{egwohdmy} = \exp\left(\beta_H \mathrm{H}_{wohdmy} + \beta_N \mathrm{N}_{wohdmy} + \beta_K \mathrm{K}_{wohdmy} + \zeta_h + \theta_{\text{weekday}} + \Gamma_{wmy} + \Phi_O + \epsilon_{egwohdmy}\right) \tag{1}$$

- w - HUC 12
- o - checklist
- O - observer ID
- h, d - hour of day, day of year
- m, y - month of year, year
- e - 1: breeding birds, 0: else
- g - grouping of bird (14 groups)

- B - observation count (and species count)
- K - km distance traveled for a checklist
- N - number of observers for a checklist
- H - hours spent birding for a checklist
- ζ_h - hour-of-day FEs (visibility/detectability)
- θ_{weekday} - weekday FEs
- Γ_{wmy} - HUC 12-month-year FEs (OUTCOME OF INTEREST)

Identification (cont.)

Fixed Effects Model

$$\text{std}(\hat{\Gamma})_{egwmy} = \beta(\text{Protected Wetland}_{wy}) + \gamma \text{g}(\text{weather}_{wmy}, \Omega) + \psi X_c + \eta_w + \delta y + \xi_{wy} \quad (2)$$

- w, m, y - as before
- c - county
- Protected Wetland - share of wetland in w protected by easements for time y i.e. $\frac{\text{area of wetland intersecting an easement}_{wmy}}{\text{area of wetland}_{wy}}$
- $\text{g}(\text{weather}_{wmy}, \Omega)$ - average temperature and precipitation
- η_w - HUC12 FEs
- X_c — controls: weather; ag land value

Shift–Share IV: Motivation

Identification Challenge

- Easement placement is endogenous: landowners, NGOs, and agencies target ecologically valuable wetlands.
- Bird abundance may directly influence protection decisions.

Exogenous Legal Shocks

- Supreme Court rulings reshaped federal wetland jurisdiction:
  - Rapanos (2006)
  - Clean Water Rule (2015)
  - Navigable Waters Protection Rule (2020)

$\Rightarrow$ These are nationwide policy shifts, plausibly unrelated to subwatershed-level bird trends.

Shift–Share Design

- *Share:* Watersheds differ in baseline predicted exposure to WOTUS jurisdiction.
- *Shift:* Interact pre-determined exposure (predicted scores) with regime timing dummies to generate spatially heterogeneous shocks to easement incentives.

Shift–Share IV: Set-up

Instrument

$$Z_{wy} = s_w^{\text{Rapanos}} \cdot D_y^{\text{CWR}} + s_w^{\text{CWR}} \cdot D_y^{\text{NWPR}}$$

$$D_y^{\text{CWR}} = \mathbf{1}\{2015 < y \leq 2020\}, \qquad D_y^{\text{NWPR}} = \mathbf{1}\{y > 2020\}.$$

First Stage

$$\text{ProtectedWetland}_{wy} = \pi Z_{wy} + \gamma g(\text{weather}_{wmy}, \Omega) + \psi X_c + \eta_w + \delta y + \mu_{wy}$$

Second Stage

$$\text{std}(\hat{\Gamma})_{egwmy} = \beta \widehat{\text{ProtectedWetland}}_{wy} + \gamma g(\text{weather}_{wmy}, \Omega) + \psi X_c + \eta_w + \delta y + \xi_{wy}$$

- w, m, y - as before
- c - county
- Protected Wetland - share of wetland in w protected by easements for year y
- $g(\text{weather}_{wmy}, \Omega)$ - average temperature and precipitation
- η_w - HUC12 FEs
- X_c — weather and land value controls

Relative Bird Abundance - All Birds (Log. Lin.)

$\hat{\Gamma}_{wmy}$: percent change in relative abundance

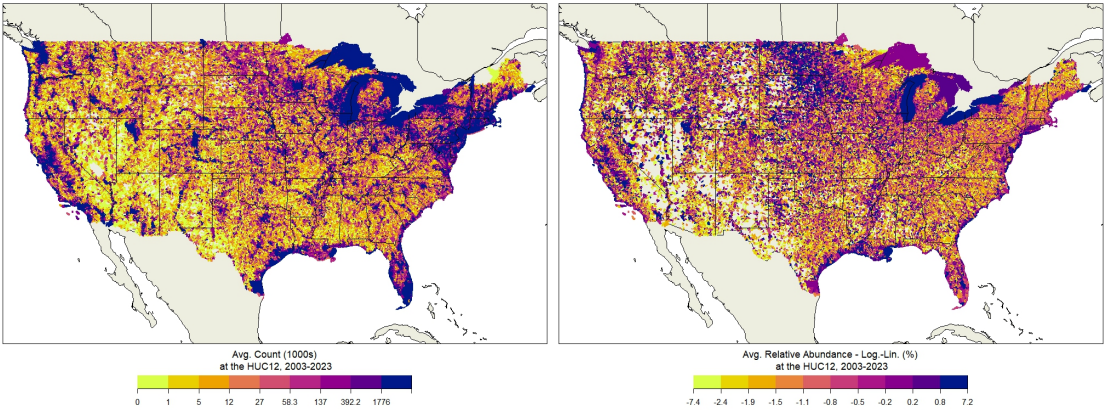


Figure: Raw Bird Count vs Change Estimated Relative Abundance (Average from 2003-2023)

Relative Species Richness - All Birds (Log. Lin.)

$\hat{\Gamma}_{wmy}$: percent change in relative richness

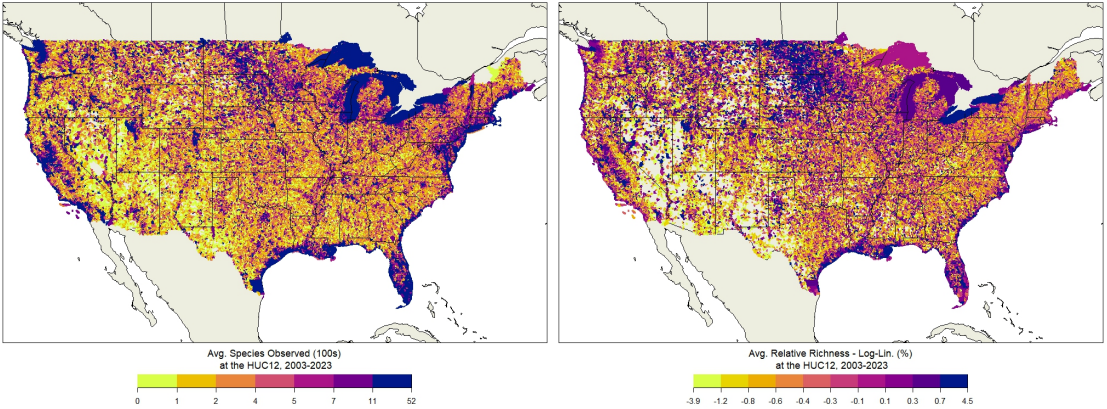


Figure: Raw Max Species Observed vs Change Estimated Relative Richness (Average from 2003-2023)

Wetland Protection and Bird Outcomes (TWFE)

	Allbirds	Waterbirds	Waterfowl	Shorebirds	Landbirds	Other
Panel A: Abundance (log, SD units)						
Protection $\hat{\beta}$	0.016	0.006	0.003	0.003	0.016	0.031
	(0.012)	(0.014)	(0.016)	(0.023)	(0.018)	(0.023)
N	11,275	6,022	7,013	4,987	11,178	6,572
Panel B: Richness (log, SD units)						
Protection $\hat{\beta}$	0.037**	0.032***	0.038**	0.012	0.021	0.052***
	(0.014)	(0.009)	(0.013)	(0.015)	(0.015)	(0.014)
N	11,332	6,092	7,088	5,034	11,243	6,687
Weather + log(county AV)				Yes		
HUC12 & Year FE				Yes		

Notes: SE clustered at HUC12. $^{\cdot}p<.10$, $^{*}p<.05$, $^{**}p<.01$, $^{***}p<.001$.

Preliminary Findings + Next Steps

- A 1-SD ($\approx$8 pp) increase in the protected share of wetland is associated with +0.03–0.05 SD higher bird species richness (0.1 SD in Rel. Richness $\approx$ 1 species); abundance effects are null.
- The richness response is concentrated among water-associated guilds, especially waterbirds and waterfowl, consistent with the ecological role of wetland protection.
- The association is strongest at the HUC12 level and remains same-signed but noisier at larger spatial scales. While the IV first stage is strong, the 2SLS estimates are still too imprecise to support a causal claim.
- **Next steps:** Probe the mechanism behind the weak IV second stage (which protection margin moves birds); Alternative IV: directly instrument easement adoption using tax incentives (federal deductions and state credits).

Q&A

Contact: tz387@cornell.edu

The IV Does Not Confirm It Yet (2SLS)

	Allbirds	Waterbirds	Waterfowl	Shorebirds	Landbirds	Other
Panel A: Abundance (log, SD units)						
Protection $\hat{\beta}$	10.06	−4.78	−3.34	−6.38	12.01	3.23
	(9.59)	(7.04)	(7.55)	(7.66)	(11.31)	(5.59)
First-stage F	97	57	67	53	97	65
N	11,275	6,022	7,013	4,987	11,178	6,572
Panel B: Richness (log, SD units)						
Protection $\hat{\beta}$	5.62	−4.38	−2.52	−4.43	4.67	8.14
	(6.06)	(6.53)	(8.29)	(6.42)	(5.63)	(8.14)
First-stage F	97	57	67	53	97	66
N	11,332	6,092	7,088	5,034	11,243	6,687
Weather + log(county AV)			Yes			
HUC12 & Year FE			Yes			

Notes: SE clustered at HUC12. $^{\cdot}p<.10$, $^{*}p<.05$, $^{**}p<.01$, $^{***}p<.001$.

Shift-Share IV: Assumptions

1. Relevance

- WOTUS regime changes shift easement enrollment, and that this shift is stronger in subwatersheds with weaker baseline federal jurisdiction (predicted scores).

$$\text{Cov}(Z_{wy}, \text{ProtectedWetland}_{wy}) \neq 0.$$

2. Exclusion Restriction

- Conditional on fixed effects, WOTUS regime shifts affect bird outcomes only by changing conservation easement enrollment incentives.

$$\mathbb{E}\left[Z_{wy}\, \xi_{wmy} \,\middle|\, g(\text{weather}_{wmy}, \Omega), \text{Rental Rate}_c, \eta_{HUC8 \times y} \right] = 0$$